Replication of a Research Claim from Sandra & Otto (2018),
from Cognition

Replication Team: Daniel Molden

Research Scientist: Melissa Kline

Action Editor: EJ Wagenmakers

Independent Reviewers
(add name below when you initiate review, comment "DONE" on your name when you finish):

Reviewer #1: Gustav Nilsonne DONE

Reviewer #2: Cristina Zogmaister DONE

Reviewer #3: [NAME]

Review Period: Feb. 6 - Feb. 11 [Updated 3-15-21]

View-only links to: Original Paper

Note to Reviewers: Going forward, we will be implementing a new preregistration form. Please reference the New Preregistration Form and the Reviewer Criteria Checklist for instructions and specific criteria to help guide your review.

Privacy Statement: Other teams are making predictions about the outcomes of many different studies, not knowing which studies have been selected for replication. As a consequence, the success of this project requires full confidentiality of this peer review process. This includes privacy about which studies have been selected for replication and all aspects of the discussion about these replication designs.

You can find the new template here

Study Information

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1. Title (required)
    - *1.1. Provide the working title of your study. It may be the same title that you submit for publication of your final manuscript, but it is not a requirement.*
    - *1.2. **Example**: Effect of sugar on brownie tastiness.*
    - *1.3. **More info**: The title should be a specific and informative description of a project. Vague titles such as 'Fruit fly preregistration plan' are not appropriate.*

TITLE: Direct Replication of a research claim from Sandra & Otto, 2018 in Cognition

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2. Authors (required)

AUTHORS:

Daniel C. Molden[1]

1 Northwestern University

3. Description (optional)
    3.1. *Please give a brief description of your study, including some background, the purpose of the of the study, or broad research questions.*
    3.2. ***Example**: Though there is strong evidence to suggest that sugar affects taste preferences, the effect has never been demonstrated in brownies. Therefore, we will measure taste preference for four different levels of sugar concentration in a standard brownie recipe to determine if the effect exists in this pastry.*
    3.3. ***More info**: The description should be no longer than the length of an abstract. It can give some context for the proposed study, but great detail is not needed here for your preregistration.*

DESCRIPTION:

The claim selected for replication from Sandra & Otto (2018) is that large reward incentives increase the amount of effort exerted by low-NFC (Need for Cognition) individuals for task-switch costs, but decrease the amount of effort exerted by high-NFC individuals. This reflects the following from the paper's abstract: "[A] low intrinsic motivation to expend effort (measured by Need for Cognition)—predicted larger increase in cognitive effort expenditure in response to monetary reward incentives, while individuals…with greater intrinsic motivation to expend effort —were less responsive to reward incentives." In the original paper, cognitive effort was operationalized in terms of performance on a switching task that required focused attention. The claim was tested using mixed-effects regression with regressors for the influence of trial type (task switch versus task repeat), reward level, experimental block, and Need for Cognition scores upon log transformed reaction times in the task-switching paradigm. This yielded a significant three-way interaction, Trial type × reward × NFC, estimate = 0.0382, se= 0.0145, p = 0.008

4. Hypotheses (required)
    - *4.1. List specific, concise, and testable hypotheses. Please state if the hypotheses are directional or non-directional. If directional, state the direction. A predicted effect is also appropriate here. If a specific interaction or moderation is important to your research, you can list that as a separate hypothesis.*
    - *4.2.* ***Example****: If taste affects preference, then mean preference indices will be higher with higher concentrations of sugar.*

HYPOTHESIS:

H (SCORE Focal hypothesis):* The study predicts a three-way interaction of trial-type, reward amount, and Need for Cognition, such that larger rewards increase performance on task-switch trials relative to task-repeat trials for low-NFC individuals, but this effect does not occur for high-NFC individuals.

H1: In a direct replication of another finding from the original study, there should be a three-way interaction of trial-type, reward amount, and an executive function measure assessed by performance on a Stroop task, such that larger rewards increase performance on task-switch trials relative to task-repeat trials for low executive function individuals (i.e. individuals showing high levels of interference on the Stroop task), but this effect does not occur for high executive function individuals (i.e., individuals showing low levels of interference on the Stroop task).

H2: In a conceptual replication of the original study, there should be a three-way interaction of trial-type, reward amount, and anticipated intrinsic motivation for the task-switching activity as measured by an adapted version of the Intrinsic Motivation Inventory (e.g., Deci, Egharari, Patrick, & Leone, 1994) administered beforehand. Analogous to the findings for NFC, larger rewards should increase performance on task-switch trials relative to task-repeat trials for low-intrinsic-motivation individuals, but this effect should not occur for high-intrinsic-motivation individuals.

Design Plan

In this section, you will be asked to describe the overall design of your study. Remember that this research plan is designed to register a single study, so if you have multiple experimental designs, please complete a separate preregistration.

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5. *Study type (required)*
    - ***5.1. Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.***
    - *5.2. Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, natural experiments, and regression discontinuity designs.*
    - *5.3. Meta-Analysis - A systematic review of published studies.*
    - *5.4. Other*

STUDY TYPE:
This study involves examining the association of observed individual differences (as assessed by self-report surveys) with differential effects of randomly assigned experimental treatments.

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6. Blinding (required)
    6.1. Blinding describes who is aware of the experimental manipulations within a study. Mark all that apply.
        6.1.1. No blinding is involved in this study.
        6.1.2. For studies that involve human subjects, they will not know the treatment group to which they have been assigned.
        6.1.3. Personnel who interact directly with the study subjects (either human or non-human subjects) will not be aware of the assigned treatments. (Commonly known as "double blind")
        6.1.4. Personnel who analyze the data collected from the study are not aware of the treatment applied to any given group.

7. Is there any additional blinding in this study?

BLINDING:
The order of the within-participants reward conditions and the between-participants stimulus set condition will be randomly determined by the experimental program (see cardsort.iqx). Experimenters directly involved in data collection will be unaware of the order condition, and participants will be unaware of the alternative order condition. Participants will experience all conditions of the within-participants manipulations, and will thus be aware of these manipulations, but will be blind to their hypothesized effects. Experimenters will also be blind to individual participants' scores on the Stroop executive function task or any of the individual difference questionnaires administered during the experimental session.

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8. Study design (required)
   - *8.1. Describe your study design. Examples include two-group, factorial, randomized block, and repeated measures. Is it a between (unpaired), within-subject (paired), or mixed design? Describe any counterbalancing required. Typical study designs for observation studies include cohort, cross sectional, and case-control studies.*
   - *8.2. Example: We have a between subjects design with 1 factor (sugar by mass) with 4 levels.*
   - *8.3. More info: This question has a variety of possible answers. The key is for a researcher to be as detailed as is necessary given the specifics of their design. Be careful to determine if every parameter has been specified in the description of the study design. There may be some overlap between this question and the following questions. That is OK, as long as sufficient detail is given in one of the areas to provide all of the requested information. For example, if the study design describes a complete factorial, 2 X 3 design and the treatments and levels are specified previously, you do not have to repeat that information.*

STUDY DESIGN:

The study design features two primary experimental manipulations: a within-participants manipulation of trial-type in a task-switching paradigm (*task-switch* vs. *task-repeat*) and a within-participants manipulation of reward amount for correct answers (*larger* vs. *smaller*) in the task-switching paradigm. As in the original study, the former manipulation will involve either switching or repeating one of two tasks: magnitude estimation (greater or less than 5) versus "even or odd" judgments of the numeral displayed on a standard playing card. The latter manipulation will involve receiving either $.05 or $.01 for each correct answer. The reward amount condition will remain consistent over a full set of trials in the task-switching paradigm. There will also be between-participants manipulations of the order of the two reward conditions (*larger-firs*t vs. *smaller-first*) and the stimulus set (*order1* vs. *order2*) assigned to each reward condition. Thus, the overall design is a 2(trial type) x 2(reward amount) x 2(reward order) x 2(stimulus set) factorial, with repeated-measures on the first two factors. The focal analysis for SCORE additionally includes a measured covariate (Need for cognition) and the analysis for Hypothesis 1 requires a second measured covariate (anticipated intrinsic motivation for the task).

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9. Randomization
    9.1. If you are doing a randomized study, how will you randomize, and at what level?
    9.2. Example: We will use block randomization, where each participant will be randomly assigned to one of the four equally sized, predetermined blocks. The random number list used to create these four blocks will be created using the web applications available at http://random.org.
    9.3. More info: Typical randomization techniques include: simple, block, stratified, and adaptive covariate randomization. If randomization is required for the study, the method should be specified here, not simply the source of random numbers.

RANDOMIZATION:
As consistent with the raw data files from the original study, for the Task-Switching task, within each block of 14 trials, seven cards marked with hearts and seven cards marked with spades will appear. Each suit will include cards numbered from 2 – 9, excluding 5. The presentation of each of these stimuli will be block-randomized. This will be done by assigning a stimulus number from 1-14 to each card and then generating random sequences of these integers using random.org. Ten different block-randomized sequences will be generated for the 10 total blocks of the first set of trials in the task-switching paradigm, and participants will then all complete the full sequence of 140 trials in the same order. After evaluating the number of *repeat* versus *switch* trials this creates (see below), manual adjustments will be made so that the number of trial types for each stimulus type as are equivalent as possible (although one of these categories will have one fewer trial because the first trial of the sequence can be neither a switch or a repeat from the trial before it). This whole procedure will then be repeated for the second set of trials in the task-switching paradigm.

The between-participants order manipulation for the within-participants reward condition will be fully counter-balanced with the two different trial sequences. The online software used to present stimulus materials will randomly assign participants to one of four Order (large-reward first vs. small-reward first) x Trial Sequence (1 vs. 2) conditions with the constraint that an equal number of participants be assigned to each condition.

As consistent with the raw data files from the original study, for the Stroop-task measure of executive function required to test Hypothesis 1, within each block of 12 trials, each color word (“red”, “blue”, “green”) will appear four times each. In three of these appearances, the word will appear in the congruent color. In the remaining instance, the word will appear in an incongruent color (“red” in a blue font, “green” in a red font, and “blue” in a green font). The presentation order of each of these stimuli will be block-randomized by the experimental program for 10 total blocks for each individual participant.

Sampling Plan

In this section we'll ask you to describe how you plan to collect samples, as well as the number of samples you plan to collect and your rationale for this decision. Please keep in mind that the data described in this section should be the actual data used for analysis, so if you are using a subset of a larger dataset, please describe the subset that will actually be used in your study.

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10. Existing data (required; multiple choice, select one option)

10.1. Preregistration is designed to make clear the distinction between confirmatory tests, specified prior to seeing the data, and exploratory analyses conducted after observing the data. Therefore, creating a research plan in which existing data will be used presents unique challenges. ***Please select the description that best describes your situation.*** *Please do not hesitate to contact us if you have questions about how to answer this question (prereg@cos.io).*

10.1.1. Registration prior to creation of data: As of the date of submission of this research plan for preregistration, the data have not yet been collected, created, or realized.

10.1.2. Registration prior to any human observation of the data: As of the date of submission, the data exist but have not yet been quantified, constructed, observed, or reported by anyone - including individuals that are not associated with the proposed study. Examples include museum specimens that have not been measured and data that have been collected by non-human collectors and are inaccessible.

10.1.3. Registration prior to accessing the data: As of the date of submission, the data exist, but have not been accessed by you or your collaborators. Commonly, this includes data that has been collected by another researcher or institution.

10.1.4. Registration prior to analysis of the data: As of the date of submission, the data exist and you have accessed it, though no analysis has been conducted related to the research plan (including calculation of summary statistics). A common situation for this scenario when a large dataset exists that is used for many different studies over time, or when a data set is randomly split into a sample for exploratory analyses, and the other section of data is reserved for later confirmatory data analysis.

10.1.5. Registration following analysis of the data: As of the date of submission, you have accessed and analyzed some of the data relevant to the research plan. This includes preliminary analysis of variables, calculation of descriptive statistics, and observation of data distributions. Please see cos.io/prereg for more information.

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11. Explanation of existing data (optional)

11.1. If you indicate that you will be using some data that already exist in this study, please describe the steps you have taken to assure that you are unaware of any patterns or summary statistics in the data. This may include an explanation of how access to the data has been limited, who has observed the data, or how you have avoided observing any analysis of the specific data you will use in your study.

*11.2. **Example**: An appropriate instance of using existing data would be collecting a sample size much larger than is required for the study, using a small portion of it to conduct exploratory analysis, and then registering one particular analysis that showed promising results. After registration, conduct the specified analysis on that part of the dataset that had not been investigated by the researcher up to that point.*

*11.3. **More info**: An appropriate instance of using existing data would be collecting a sample size much larger than is required for the study, using a small portion of it to conduct exploratory analysis, and then registering one particular analysis that showed promising results. After registration, conduct the specified analysis on that part of the dataset that had not been investigated by the researcher up to that point.*

EXPLANATION OF EXISTING DATA: N/A

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12. Data collection procedures (required)

12.1. Please describe the process by which you will collect your data. If you are using human subjects, this should include the population from which you obtain subjects, recruitment efforts, payment for participation, how subjects will be selected for eligibility from the initial pool (e.g. inclusion and exclusion rules), and your study timeline. For studies that donÍt include human subjects, include information about how you will collect samples, duration of data gathering efforts, source or location of samples, or batch numbers you will use.

*12.2. **Example**: Participants will be recruited through advertisements at local pastry shops. Participants will be paid $10 for agreeing to participate (raised to $30 if our sample size is not reached within 15 days of beginning recruitment). Participants must be at least 18 years old and be able to eat the ingredients of the pastries.*

*12.3. **More information**: The answer to this question requires a specific set of instructions so that another person could repeat the data collection procedures and recreate the study population. Alternatively, if the study population would be unable to be reproduced because it relies on a specific set of circumstances unlikely to be recreated (e.g., a community of people from a specific time and location), the criteria and methods for creating the group and the rationale for this unique set of subjects should be clear.*

DATA COLLECTION PROCEDURES:

Participants will be recruited from the online subject pool managed by the website prolific.co and complete all tasks and materials online using Qualtrics and the Inquisit Web software (https://www.millisecond.com/products/inquisit6/weboverview.aspx) . Although the original Sandra & Otto (2018) does not report any demographics for their sample, participants for this study were recruited from a university subject pool and community in Canada. Participants for the current study will therefore be limited to individuals born and living in North America who report English as their first language. In addition, because the original sample was likely mostly comprised of university students, only participants between the ages of 18 and 40 will be recruited. Participants will first read the consent form and indicate informed consent. They will then report their age and gender, and complete the Need for Cognition and BIS/BAS individual difference measures (see Qualtrics_survey.docx). One attention-check item that instructs participants to select a particular response will be embedded in the Need for Cognition questionnaire (question #16; see Qualtrics_survey.docx) and one will be embedded in the BIS/BAS questionnaire (question #7; see Qualtrics_survey.docx). These items will be used to exclude participants who are not carefully reading these questionnaires (see section 22 below).

Participants will next complete a computerized Stroop task run through Inquisit Web (stroop_Sanders.iqx), which will require them to identify, as quickly and as accurately as possible, which one of three colors the word on the screen is presented: red, green, or blue, by pressing one of three keys (*j*, *k*, and *l* respectively) using their dominant hand while ignoring the meaning of the

word. Before starting with the task, participants will first complete a short practice block to get them accustomed to the task. The main Stroop task will consist of 120 trials, 30 incongruent and 90 congruent. On each trial, the participant will see the stimulus (a color word sized 100 × 350 pixels) 500 ms after the onset of the trial. The participant will then have 1.5 s to make a response. No feed-back will be provided. RGB color codes (255, 0, 0), (0, 255, 0), (0, 0, 255) will be used for red, green, and blue respectively.

After performing the Stroop, participants will be randomly assigned to one of four reward-order x trial-set between-participants conditions (see section 9) by the card sort Inquisit Web program (cardsort.iqx) and informed that they will complete another computerized task. They will read the full instructions for a modified card-choice version of the task used by Kool et al. (2010). Participants will be presented with a picture of a common playing card of one of two suits: Spades (black) and Hearts (red). Each card will contain a red or black number in the center (color corresponding to the suit of the card) indicating its value; the cards will vary from 2 to 9, omitting 5. The number stimulus will be presented in the center, the top-left, and the bottom-right corners of a white “card” in a 40-point Times New Roman font in either red or black. Upon presentation of a red card the participant will use one of the two response keys (*f* or *j*, respectively) to indicate whether the card is smaller or larger than 5 (the *magnitude* task). Upon presentation of a black card, the participant will use one of the two response keys (*f* or *j*, respectively) to indicate whether the number on the card is odd or even (the *parity* task).

Participants will be presented with the stimulus (card) 100 ms after the onset of the trial; they will have 3s to respond and will be given feedback on the correctness of their response. If the participant takes too long to answer, the trial will be automatically counted as a miss and the next card will be shown. Consistent with the raw data files from the original study, after reading all of the instructions, participants will complete a practice block of 14 trials.

Following the practice block, participants will complete eight items from an Intrinsic Motivation Inventory adapted from the *activity perception* questionnaire used Deci, Egharari, Patrick, and Leone (1994). These items will be adapted from the *interest*/*enjoyment* subscale and will be worded to ask participants to forecast their experience on the upcoming task (see Intrinsic Motivation Inventory.docx and cardsort.iqx). Finally, participants will receive instructions for a performance-contingent cash bonus. Based on their randomized condition code, they will either be informed that they will receive $.05 (*larger reward*) or $.01 (*smaller reward*) for each correct answer, and will complete the first set of 140 trials of the main task-switching paradigm. Following this, participants will have an 8-min music break, featuring a selection of classical pieces by Bach (rest_video.mp4), and then complete the second set of 140 trials of the task-switching paradigm in the remaining reward condition. Participants will receive performance feedback after each trial that is specific to the reward condition they are currently completing.

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13. Sample size (required)

13.1. Describe the sample size of your study. How many units will be analyzed in the study? This could be the number of people, birds, classrooms, plots, interactions, or countries included. If the units are not individuals, then describe the size requirements for each unit. If you are using a clustered or multilevel design, how many units are you collecting at each level of the analysis?

13.2. ***Example****: Our target sample size is 280 participants. We will attempt to recruit up to 320, assuming that not all will complete the total task.*

13.3. ***More information****: For some studies, this will simply be the number of samples or the number of clusters. For others, this could be an expected range, minimum, or maximum number.*

SAMPLE SIZE:

The initial target sample size is **122 participants**. If a statistically significant effect at $p < .05$ is not observed after the first round of data collection, a second round will begin. The second round of data collection will sample an additional **152 participants** for a pooled sample of **274 participants.**

Because of the ~15% attrition rate in the original study due to inattention on the cognitive tasks, to achieve the target analytic sample, the target sample size for recruitment is **144 participants**. If necessary, the second round of data collection will sample an additional **180 participants** for a pooled recruited sample size of **324 participants**.

14. Sample size rationale (optional)
    - *14.1. This could include a power analysis or an arbitrary constraint such as time, money, or personnel.*
    - *14.2.* ***Example****: We used the software program G*Power to conduct a power analysis. Our goal was to obtain .95 power to detect a medium effect size of .25 at the standard .05 alpha error probability.*
    - *14.3.* ***More information****: This gives you an opportunity to specifically state how the sample size will be determined. A wide range of possible answers is acceptable; remember that transparency is more important than principled justifications. If you state any reason for a sample size upfront, it is better than stating no reason and leaving the reader to "fill in the blanks." Acceptable rationales include: a power analysis, an arbitrary number of subjects, or a number based on time or monetary constraints.*

SAMPLE SIZE RATIONALE:

Power calculations were done in accordance with the guidelines of the Social Sciences Replication Project (SSRP). The first round of data collection achieves 90% power to detect 75% of the original effect size. The pooled sample, if necessary after testing the effect on the first round of data, achieves 90% power to detect 50% of the original effect size.

The power analysis for this replication is stored in the OSF project associated with this replication attempt, in the power analysis component. A view-only link to this component is provided here: https://osf.io/ekz9r/?view_only=c6da7cbee24f49ccbd66f7176b548ae0.

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15. Stopping rule (optional)

15.1. If your data collection procedures do not give you full control over your exact sample size, specify how you will decide when to terminate your data collection.

*15.2. **Example**: We will post participant sign-up slots by week on the preceding Friday night, with 20 spots posted per week. We will post 20 new slots each week if, on that Friday night, we are below 320 participants.*

*15.3. **More information**: You may specify a stopping rule based on p-values only in the specific case of sequential analyses with pre-specified checkpoints, alphas levels, and stopping rules. Unacceptable rationales include stopping based on p-values if checkpoints and stopping rules are not specified. If you have control over your sample size, then including a stopping rule is not necessary, though it must be clear in this question or a previous question how an exact sample size is attained.*

STOPPING RULE:

The planned sample size is **144**. After achieving that sample, planned analyses will be run. If a significant effect is found, sampling stops. If a significant effect is not found, a second round of data collection will collect data from **180** additional participants, for a pooled sample of **324**. Sampling will stop after the second round of data collection regardless of a significant effect.

Variables

In this section you can describe all variables (both manipulated and measured variables) that will later be used in your confirmatory analysis plan. In your analysis plan, you will have the opportunity to describe how each variable will be used. If you have variables which you are measuring for exploratory analyses, you are not required to list them, though you are permitted to do so.

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16. Manipulated variables
    - *16.1. Describe all variables you plan to manipulate and the levels or treatment arms of each variable. This is not applicable to any observational study.*
    - *16.2.* ***Example:*** *We manipulated the percentage of sugar by mass added to brownies. The four levels of this categorical variable are: 15%, 20%, 25%, or 40% cane sugar by mass.*
    - *16.3.* ***More information****: For any experimental manipulation, you should give a precise definition of each manipulated variable. This must include a precise description of the levels at which each variable will be set, or a specific definition for each categorical treatment. For example, "loud or quiet," should instead give either a precise decibel level or a means of recreating each level. 'Presence/absence' or 'positive/negative' is an acceptable description if the variable is precisely described.*

MANIPULATED VARIABLES:
There are four manipulated variables. The first will be whether participants will perform the same or a different task in consecutive trials on the task-switching paradigm. If participants either perform two consecutive magnitude judgments when viewing a red card, or perform two consecutive parity judgments when viewing a black card, their responses for the second judgment will be scored as a *task-repeat* response. If participants either perform a parity judgment when viewing a black card immediately following a magnitude judgment when viewing a red card or a magnitude judgment when viewing a red card following a parity judgment when viewing a black card, their responses for the second judgment will be scored as a *task-switch* response. This manipulation is within-participants and equal numbers of both types of responses will be required in each of 14 blocks of the task-switching task, as detailed above.

The second manipulated variable will be whether participants complete the task-switch paradigm with the incentive of either $.05 for each correct answer (*larger reward* condition) or $.01 for each correct answer (*smaller reward* condition). This manipulation is within-participants and participants will complete separate full sets of 140 task-switch trials under both reward conditions, as detailed above.

The third and fourth manipulated variables will be whether participants complete the larger-reward set of task-switch trials first and the smaller-reward set after a break, and with which reward condition the first or second block-randomized sequence of trials will be paired (see above). This 2(reward) x 2(order) manipulation is between participants and will ensure a fully counterbalanced design for the reward-order and paired-sequence manipulations.

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17. Measured variables (required)

17.1. Describe each variable that you will measure. This will include outcome measures, as well as any predictors or covariates that you will measure. You do not need to include any variables that you plan on collecting if they are not going to be included in the confirmatory analyses of this study.

*17.2. **Example**: The single outcome variable will be the perceived tastiness of the single brownie each participant will eat. We will measure this by asking participants 'How much did you enjoy eating the brownie' (on a scale of 1-7, 1 being 'not at all', 7 being 'a great deal') and 'How good did the brownie taste' (on a scale of 1-7, 1 being 'very bad', 7 being 'very good').*

*17.3. **More information**: Observational studies and meta-analyses will include only measured variables. As with the previous questions, the answers here must be precise. For example, 'intelligence,' 'accuracy,' 'aggression,' and 'color' are too vague. Acceptable alternatives could be 'IQ as measured by Wechsler Adult Intelligence Scale' 'percent correct,' 'number of threat displays,' and 'percent reflectance at 400 nm.'*

MEASURED VARIABLES:

The first measured independent variable will be participants' executive function, as assessed by their performance on the Stroop task. Executive function will be calculated as the difference between standardized RTs on incongruent and congruent trials. The second measured independent variable will be participants' Need for Cognition, as assessed by their responses on the 18-item scale created by Cacioppo, Petty, and Kao (1984; see ncogscale.pdf). The third measured independent variable will be participants' scores on the BIS/BAS inventory by Carver & White (1994; see BISBAS.pdf). The fourth measured independent variable will be participants' anticipated intrinsic motivation for the task-switching task, as assessed by the seven-item *interest*/*enjoyment* subscale on the *activity perception* questionnaire created by Deci, Egharari, Patrick, and Leone (1994; see IMI.pdf), with the items reworded to be in the future tense concerning the upcoming activity.

The first measured dependent variable will be participants' reaction times (RTs) on each trial of the task-switching paradigm. The second measured dependent variable will be whether or not participants responded correctly on each trial of the task-switching paradigm. An additional covariate that will be measured for inclusion in the primary analyses will be the block number of the trials of the task-switch paradigm.

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18. Indices

18.1. If any measurements are going to be combined into an index (or even a mean), what measures will you use and how will they be combined? Include either a formula or a precise description of your method. If your are using a more complicated statistical method to combine measures (e.g. a factor analysis), you can note that here but describe the exact method in the analysis plan section.

***18.2. Example**: We will take the mean of the two questions above to create a single measure of 'brownie enjoyment.'*

*18.3. **More information**: If you are using multiple pieces of data to construct a single variable, how will this occur? Both the data that are included and the formula or weights for each measure must be specified. Standard summary statistics, such as "means" do not require a formula, though more complicated indices require either the exact formula or, if it is an established index in the field, the index must be unambiguously defined. For example, "biodiversity index" is too broad, whereas "Shannon's biodiversity index" is appropriate.*

INDICES:

The index of each participant's executive function score as measured by the Stroop Task will be calculated as the difference between standardized mean RTs on the incongruent and standardized mean RTs congruent trials. The indices of Need for Cognition, BIS/BAS, and anticipated intrinsic motivation will be calculated by summing participants' responses to the relevant items, reverse-scored where appropriate, as detailed in the documents describing each questionnaire. Separate mean accuracy and mean RT indices will be calculated for participants' responses on task-repeat versus task-switch trials in each block of both sets of trials on the task-switch paradigm. A *block number* index will also be created for the task-switch paradigm by consecutively numbering each set of 14 trials on this task (see the Randomization section for further details).

Analysis Plan

You may describe one or more confirmatory analysis in this preregistration. Please remember that all analyses specified below must be reported in the final article, and any additional analyses must be noted as exploratory or hypothesis generating.

A confirmatory analysis plan must state up front which variables are predictors (independent) and which are the outcomes (dependent), otherwise it is an exploratory analysis. You are allowed to describe any exploratory work here, but a clear confirmatory analysis is required.

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19. Statistical models (required)
    - *19.1. What statistical model will you use to test each hypothesis? Please include the type of model (e.g. ANOVA, multiple regression, SEM, etc) and the specification of the model (this includes each variable that will be included as predictors, outcomes, or covariates). Please specify any interactions, subgroup analyses, pairwise or complex contrasts, or follow-up tests from omnibus tests. If you plan on using any positive controls, negative controls, or manipulation checks you may mention that here. Remember that any test not included here must be noted as an exploratory test in your final article.*
    - ***19.2. Example**: We will use a one-way between subjects ANOVA to analyze our results. The manipulated, categorical independent variable is 'sugar' whereas the dependent variable is our taste index.*
    - *19.3. **More information**: This is perhaps the most important and most complicated question within the preregistration. As with all of the other questions, the key is to provide a specific recipe for analyzing the collected data. Ask yourself: is enough detail provided to run the same analysis again with the information provided by the user? Be aware for instances where the statistical models appear specific, but actually leave openings for the precise test. See the following examples:*
        - *19.3.1.1. If someone specifies a 2x3 ANOVA with both factors within subjects, there is still flexibility with the various types of ANOVAs that could be run. Either a repeated measures ANOVA (RMANOVA) or a multivariate ANOVA (MANOVA) could be used for that design, which are two different tests.*
        - *19.3.1.2. If you are going to perform a sequential analysis and check after 50, 100, and 150 samples, you must also specify the p-values you'll test against at those three points.*

STATISTICAL MODELS:
The primary dependent variables of mean accuracy and RT on the task-switch task will be analyzed using multi-level regression models specified with the lme4 package

(Pinheiro & Bates, 2000) for the R programming language. The lmerTest package (Kuznetsova, Brockhoff, & Christensen, 2017), which uses the Satterthwaite's method to calculate degrees of freedom, will be used to generate *p* values. For the primary replication analyses, the within-participants reward (*rewardlevel*; -1 = smaller reward, 1 = larger reward) and trial type (*trial*; -1 = task-repeat, 1 = task-switch) conditions will be contrast coded and the individual difference moderator of Need for Cognition (*NFC*) will be entered as a continuous variable. A continuous covariate that indexes 10 separate blocks of 14 trials in the task-switching paradigm will also be contrast-coded (*blocknumber*; -5, -4, -3, -2, -1, 1, 2, 3, 4, 5). The analysis will have a two-level structure with reward and trial manipulated, and block number measured, at Level 1 and with individual differences in Need for Cognition measured at Level 2.

The primary replication analysis examining the three-way interaction of Need for Cognition with reaction times on the task-switching paradigm will first remove all incorrect RT scores and then log-transform this variable as specified in the original paper (see below). The transformed scores will then be analyzed with a regression model specified by the following code:

- lmer(logRT ~ NFC * trial * rewardlevel + blocknumber + (1|SubjectID))

SubjectID will represent the numeric code assigned to each individual participant. If a significant ($p < .05$) three-way interaction is found, follow-up analyses will then be conducted to examine the trial x rewardlevel interactions for participants high versus low in the Need for Cognition by creating new NFC indices centered at +1SD (*HiNFC*) and -1SD (*LoNFC)* from the sample mean. These regression models will be specified by the following code:

- lmer(logRT ~ HiNFC * trial * rewardlevel + blocknumber + (1|SubjectID))
- lmer(logRT ~ LoNFC * trial * rewardlevel + blocknumber + (1|SubjectID))

Several additional analyses will repeat other analyses featured in the original paper. The association of Need for Cognition with accuracy scores on the task-switching task (coded 0 = incorrect, 1 = correct) will be examined with a binary logistic regression model specified by the following code:

- glmer(accuracy ~ NFC * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

If a significant ($p < .05$) three-way interaction is found, follow-up analyses will then be conducted to examine the trial x rewardlevel interactions for participants high versus low in the Need for Cognition with the following code:

- glmer(accuracy ~ HiNFC * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")
- glmer(accuracy ~Lo NFC * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

Analyses examining the association of executive function with accuracy and reaction time to test Hypothesis 1 will follow the exact same procedures as above, replacing the NFC variable with the executive function index calculated from the Stroop task (*Stroopindex*; see details above).

- lmer(logRT ~ Stroopindex * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ Stroopindex * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

If a significant ($p < .05$) three-way interactions are found, follow-up analyses will then be conducted to examine the trial x rewardlevel interactions for participants high versus low in executive function by creating new indices centered at +1SD (*HiStroopindex*) and -1SD (*LoStroopindex)* from the sample mean. These regression models will be specified by the following code:

- lmer(logRT ~ HiStroopindex * trial * rewardlevel + blocknumber + (1|SubjectID))
- lmer(logRT ~ LoStroopindex * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ HiStroopindex * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")
- glmer(accuracy ~ LoStroopindex * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

The original paper featured analyses that repeated those above controlling for reward sensitivity as measured by the BAS index as an additional covariate. However, these reported analyses did not control for interactions between BAS and the within-participants manipulations (see Yzerbyt, Muller, & Judd, 2004). To improve these covariate analyses, the present study will add the additional interaction terms and will be conducted with regression models specified by the following code:

- lmer(logRT ~ NFC * trial * rewardlevel + blocknumber + BAS * trial * rewardlevel + (1|SubjectID))
- glmer(accuracy ~ NFC * trial * rewardlevel + blocknumber + BAS * trial * rewardlevel + (1|SubjectID), family="binomial")
- lmer(logRT ~ Stroopindex * trial * rewardlevel + blocknumber + BAS * trial * rewardlevel + (1|SubjectID))
- glmer(accuracy ~ Stroopindex * trial * rewardlevel + blocknumber + BAS * trial * rewardlevel + (1|SubjectID), family="binomial")

To test Hypothesis 2, the primary regression analyses will be repeated replacing the Need for Cognition index with the anticipated intrinsic motivation index (*IMI*) as a moderator. These models will be specified by the following code:

- lmer(logRT ~ IMI * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ IMI * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

If a significant ($p < .05$) three-way interactions are found, follow-up analyses will then be conducted to examine the trial x rewardlevel interactions for participants high versus low in anticipated intrinsic motivation by creating new indices centered at +1SD (*HiIMI*) and -1SD (*LoIMI)* from the sample mean. These regression models will be specified by the following code:

- lmer(logRT ~ HiIMI * trial * rewardlevel + blocknumber + (1|SubjectID))
- lmer(logRT ~ LoIMI * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ HiIMI * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")
- glmer(accuracy ~ LoIMI * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

20. Transformations

20.1. If you plan on transforming, centering, recoding the data, or will require a coding scheme for categorical variables, please describe that process.

*20.2. **Example**: The "Effect of sugar on brownie tastiness" does not require any additional transformations. However, if it were using a regression analysis and each level of sweet had been categorically described (e.g. not sweet, somewhat sweet, sweet, and very sweet), 'sweet' could be dummy coded with 'not sweet' as the reference category.*

*20.3. **More information**: If any categorical predictors are included in a regression, indicate how those variables will be coded (e.g. dummy coding, summation coding, etc.) and what the reference category will be.*

TRANSFORMATIONS:

As in the original paper, RT analyses will only be performed on correct answers, and RTs for incorrect answers will be dropped from all analyses. The remaining RTs will be transformed with a natural log before analyses are conducted (see Data_cleaning_syntax.docx).

All categorical variables representing experimental conditions will be coded in the manner described in the previous section. All continuous moderator variables will be standardized before being entered into the regression models (see Data_cleaning_syntax.docx).

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21. Inference criteria (optional)

21.1. What criteria will you use to make inferences? Please describe the information youÍll use (e.g. p-values, bayes factors, specific model fit indices), as well as cut-off criterion, where appropriate. Will you be using one or two tailed tests for each of your analyses? If you are comparing multiple conditions or testing multiple hypotheses, will you account for this?

*21.2. **Example**: We will use the standard p<.05 criteria for determining if the ANOVA and the post hoc test suggest that the results are significantly different from those expected if the null hypothesis were correct. The post-hoc Tukey-Kramer test adjusts for multiple comparisons.*

*21.3. **More information:** P-values, confidence intervals, and effect sizes are standard means for making an inference, and any level is acceptable, though some criteria must be specified in this or previous fields. Bayesian analyses should specify a Bayes factor or a credible interval. If you are selecting models, then how will you determine the relative quality of each? In regards to multiple comparisons, this is a question with few "wrong" answers. In other words, transparency is more important than any specific method of controlling the false discovery rate or false error rate. One may state an intention to report all tests conducted or one may conduct a specific correction procedure; either strategy is acceptable.*

INFERENCE CRITERIA:

Criteria for a successful replication attempt for the SCORE project is a statistically significant effect (alpha = .05, two tailed) in the same pattern as the original study. For this replication attempt, this criteria is met for the focal SCORE test (H*) by a significant three-way interaction of trial-type, reward amount, and Need for Cognition, such that larger rewards lead to smaller differences in reaction times between task-switch and task-repeat trials as compared to smaller rewards for low-NFC individuals, but this effect does not occur for high-NFC individuals.

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22. Data exclusion

22.1. How will you determine what data or samples, if any, to exclude from your analyses? How will outliers be handled? Will you use any awareness check?

22.2. ***Example****: No checks will be performed to determine eligibility for inclusion besides verification that each subject answered each of the three tastiness indices. Outliers will be included in the analysis.*

22.3. ***More information****: Any rule for excluding a particular set of data is acceptable. One may describe rules for excluding a participant or for identifying outlier data.*

DATA EXCLUSIONS:

Consistent with the original paper, participants who score lower than 80% accuracy on the Stroop task or the task-switching paradigm or who do not respond within the maximum allotted time for 15 or more trials in either task will not be included in any analyses. Participants who fail either one of the attention-check items embedded in the Need for Cognition and BIS/BAS questionnaires by not selecting the specified option will also be excluded. All other participants with complete responses on all tasks and the Need for Cognition questionnaire will be included in the analyses.

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23. Missing data

23.1. How will you deal with incomplete or missing data?

*23.2. **Example**: If a subject does not complete any of the three indices of tastiness, that subject will not be included in the analysis.*

*23.3. **More information**: Any relevant explanation is acceptable. As a final reminder, remember that the final analysis must follow the specified plan, and deviations must be either strongly justified or included as a separate, exploratory analysis.*

MISSING DATA:

Consistent with the original paper, participants who do not complete all of the items on the Need for Cognition questionnaire will not be included in any analyses. Participants who do not complete all of the items on the BIS/BAS or anticipated intrinsic motivation questionnaires will not be included in the analyses involving these moderators.

Also consistent with the original paper, participants who miss 15 or more response deadlines during the Stoop task or during either of the two sets of trials on task-switching paradigm will not be included in any analyses.

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24. Exploratory analysis (optional)

24.1. If you plan to explore your data set to look for unexpected differences or relationships, you may describe those tests here. An exploratory test is any test where a prediction is not made up front, or there are multiple possible tests that you are going to use. A statistically significant finding in an exploratory test is a great way to form a new confirmatory hypothesis, which could be registered at a later time.

*24.2. **Example**: We expect that certain demographic traits may be related to taste preferences. Therefore, we will look for relationships between demographic variables (age, gender, income, and marital status) and the primary outcome measures of taste preferences.*

EXPLORATORY ANALYSIS:

If significant three-way interactions are found for either accuracy or reactions times in both the Need for Cognition analyses and the anticipated intrinsic motivation analyses described above, follow-up analyses will be conducted in which both effects are examined controlling for the other:

- lmer(logRT ~ NFC * trial * rewardlevel + IMI * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ NFC * trial * rewardlevel + IMI * trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

These analyses will explore whether either one of these two individual difference measures is a better measure of the proposed intrinsic motivation effect in the original paper. It is possible that specifically asking people about their anticipated interest in the upcoming task better captures this construct than the more general measure of Need for Cognition.

If a significant ($p < .05$) three-way interactions are found for either moderator, follow-up analyses will then be conducted to examine the trial x rewardlevel interactions for participants scoring high versus low on that variable using the indices describe above. These regression models will be specified by the following code:

- lmer(logRT ~ [HiIMI or HiNFC] * trial * rewardlevel + blocknumber + (1|SubjectID))
- lmer(logRT ~ [LoIMI o LoNFC] * trial * rewardlevel + blocknumber + (1|SubjectID))
- glmer(accuracy ~ [HiIMI or HiNFC]* trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")
- glmer(accuracy ~ [LoIMI o LoNFC]* trial * rewardlevel + blocknumber + (1|SubjectID), family="binomial")

Other

25. Deviations from the original study (required)
    25.1. Please describe any known deviations from the original study. Include discussion of whether and how these deviations might impact the results of this replication/reproduction attempt and its comparison to the original study.
26. Other (optional)
    26.1. If there is any additional information that you feel needs to be included in your preregistration, please enter it here. Literature cited, disclosures of any related work such as replications or work that uses the same data, or other context that will be helpful for future readers would be appropriate here.

DEVIATIONS AND OTHER INFORMATION:
The primary deviation of the present protocol is that all data collection will be conducted online rather than in-person due to COVID-19 protocols. Every effort will be made to ensure online participants have the same experience as in-person participants would. Inquisit Web software (https://www.millisecond.com/products/inquisit6/weboverview.aspx) will be used to ensure precision millisecond timing is still possible without lag from internet connections. Participants recruited from prolific.co will also be informed ahead of time that they will be excluded from the study and risk a loss of compensation by the terms of Prolific's data quality policy, if they do not attend sufficiently to the study.

Another deviation is that the Barratt Impulsiveness and GAD measures will not be administered as they were in the original study, because no analyses featuring these measures were reported. The elimination of these measures is also unlikely to impact this replication attempt.

A more significant change to the original protocol is that following the practice block of both sets of trials in the task-switch paradigm, participants will complete eight items directly measuring their anticipated intrinsic motivation (Deci et al., 1994) for the upcoming task. They will then proceed to the main task. Although this has more potential to influence the results of the replication attempt, it still seems unlikely to have an impact because similar questions were presented in close proximity to the task-switching paradigm in the original study when the Need for Cognition questionnaire was administered. Furthermore, the potential benefit of adding these new questions is that they may represent a more direct measure of the psychological construct for which the original authors were using the Need for Cognition questionnaire as a proxy.

Still another deviation for the original study is instead of presenting the trials of the task-switching paradigm in a different block-randomized fashion for each participant, the present protocol will create a single block-randomized order of 140 trials for each of the two sets of trials

and then present this same order to all participants. This is unlikely to impact this replication attempt and may again represent an improvement on the original study in that all participants will now experience the exact same number of task-switch and task-repeat trials in the exact same order. This change will also facilitate the scoring of the separate accuracy and RT indices for each type of trial and speed the analyses.

Finally, following the first set of task-switching task trials, participants will not complete the unspecified, irrelevant filler task featured in the original study. They will also have an 8-minute music break featuring a selection of classical pieces by Bach (rest_video.mp4) in place of the 8-minute break featuring unspecified music in the original. This is unlikely to impact this replication attempt.